Product Name: Multi-Protocol Industrial Converter (Transparent Transmission Series) **Model Name: CONV-WiFi Series** (including UART-WiFi-001, SPI-WiFi-002, CAN-WiFi-003, RS485-WiFi-004)

Version: V1.0

1. Product Introduction

The **CONV-WiFi Series** is an industrial-grade CAN/Serial-to-WiFi Bridge based on the **ESP32-C3-MINI-1-N4** module. It is designed to function as a bidirectional data pipeline, providing seamless Transparent Data Transmission between field-side MCU interfaces (such as CAN, SPI, RS485, and UART) and a remote server . In this mode, the device ensures that the original data payload remains completely unmodified as it is encapsulated and forwarded between the local physical interface and the server.

- **Upstream:** Data received from the MCU via the local interface (e.g., CAN bus) is packaged and transmitted to the server via Wi-Fi without any modification to the data payload.
- **Downstream:** Command data issued by the server is delivered via Wi-Fi and transparently forwarded to the corresponding MCU interface, ensuring the original data format is preserved.
- MCU interfaces has 3.3v and 5v With CONV-UART/RS485/SPI, MCU CAN ONLY has 5V power supply with CONV-CAN.

2. Technical Specifications

- **Main Controller:** ESP32-C3-MINI-1-N4 (RISC-V 32-bit MCU up to 160 MHz)
- **Highest Internal Frequency:** 160 MHz
- **Power Supply:**
  - **Flexible Input Options: Supports dual-voltage input modes to match your MCU system voltage.**
  - **For 5V Systems:** Connect 5.0V DC to the 5V pin or USB (J0_3). The onboard LM1117-3.3V LDO will regulate this to the required 3.3V for the system.
  - **For 3.3V Systems:** Connect 3.3V DC directly to the 3V3 pin.
  - **Safety Warning**: **Use only the provided 5.0V power adapter (Model: VEL05US050-US-JA)** when powering via the 5V/USB interface. Ensure a stable power source (>1.0 A recommended) to support peak Wi-Fi and CAN activity.

    - **Current Requirement:** Ensure a stable power source capable of providing at least **1.0 A** (recommended) to support peak Wi-Fi transmission and CAN bus activity.
    - **Operating Voltage Range:** 4.5 V to 5.5 V DC. **Do not exceed 5.5 V** to prevent excessive heat or damage to the onboard regulator.
- **Interfaces:**
    - 1x UART (TTL)
    - 1x SPI Interface
    - 1x CAN 2.0B (Terminal Block)
    - 1x RS485 (Standard industrial terminal)
- **Transmission Mode:** Transparent Transmission
- **Operating Temperature:** -40 °C to +85 °C (standard variant) or -40 °C to +105 °C (high-temp variant).
- **Dimensions**: 31 mm × 40 mm × 10 mm (Base board dimensions; overall height may vary based on terminal block and component extensions).
- **Wi-Fi**:

  • IEEE 802.11b/g/n-compliant

  • Center frequency range of operating channel: 2412 ~ 2484 MHz

  • Supports 20 MHz, 40 MHz bandwidth in 2.4 GHz band

  • 1T1R mode with data rate up to 150 Mbps

  • Ultra-Low-Power SoC with RISC-V Single-Core CPU
- *Converter* Models :

  - UART-WiFi-001

  - SPI-WiFi-002

  - CAN-WiFi-003

  - RS485-WiFi-004
- FCC Compliance Statement
  - Contains FCC ID:** 2AC7Z-ESP32C3MINI1
  - Grant Date:** May 06, 2022
  - Modular Type:** Single Modular

3. Hardware Connection

- **Power Connection:**
    - **Voltage Matching**: Select the power input based on your target MCU's system voltage.
    - **3.3V System**: Connect the 3.3V output from your MCU system to the **3V3 pin**.
    - **5V System**: Connect the 5V output from your MCU system to the **5V pin**. The internal LDO will handle the regulation to 3.3V.
    - **Grounding**: Always ensure the **GND** pin is connected to your system's common ground.
- **UART:** Native support from ESP32-C3; also, has extended with level shifters for 5V tolerance, connect TX/RX/GND to target device, Baud rate: 300–115200, 8N1.
- **SPI:** Native support from ESP32-C3; also, has extended with level shifters for 5V tolerance, connect MOSI/MISO/SCK/CS/REQ to target device, CONV-SPI only works as slave mode, Max speed: 10 MHz .
- **RS485:** Connect A+ and B- to the corresponding RS485 bus target device, it has 2 interface, 300–115200, 8N1.
- **CAN:** Connect CAN_H and CAN_L to target device, it has 2 interface, Speed: 25 kbps – 1 Mbps .

  Note: An external 120Ω termination resistor is required at the end of the CAN bus cable for stable communication.

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4. FCC Compliance Statement

Note: This device complies with **Part 15 of the FCC Rules**. Operation is subject to the following two conditions:

1. This device may not cause harmful interference, and
2. This device must accept any interference received, including interference that may cause undesired operation.

Warning: Changes or modifications to this unit not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

Class B Device Statement: *Note: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to* *provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation.*

If this equipment does cause harmful interference to radio or television reception, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

RF Exposure Warning (Regarding the wireless functionality of the ESP32-C3): This equipment must be installed and operated in accordance with provided instructions and the antenna(s) used for this transmitter must be installed to provide a separation distance of at least 20 cm from all persons.

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5. Canada ICES-003 Statement

This Class B digital apparatus complies with Canadian ICES-003. Cet appareil numérique de la classe B est conforme à la norme NMB-003 du Canada. **CAN ICES-3 (B) / NMB-3 (B)**

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6. Contact Information

Manufacturer: WNetTech

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All documents: https://github.com/wnettech/compliance